Plant or weather? What open ivy data says about reading plant electrical signals

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In four outdoor ivy plants, weather and time of day explain under 3% of the electrical signal's level. But the signal's variability roughly doubles by day, is shared across plants (R^2 0.43 from the other three plants), and is the only part that still works on a plant a model has never seen.

Why this question

A probe that reads a plant's electrical signal only helps a grower if it says something the weather station does not. So the first question for any reading is how much of it is just weather and the day-night cycle, and how much is the plant itself.

That split matters for two of Electric Plant's launch claims:

- **"A couple of probes per acre."** This works if the useful signal is mostly shared across trees. If it is mostly tree-specific, coverage needs more probes or smarter placement.
- **A model that works on species it never saw.** What transfers to a new plant has to be the part of the signal that behaves the same way across plants, not each plant's own baseline.

No public data exists for orchard trees, so this uses the best open outdoor dataset there is and treats it as a test of method, not of Electric Plant's hardware.

The data

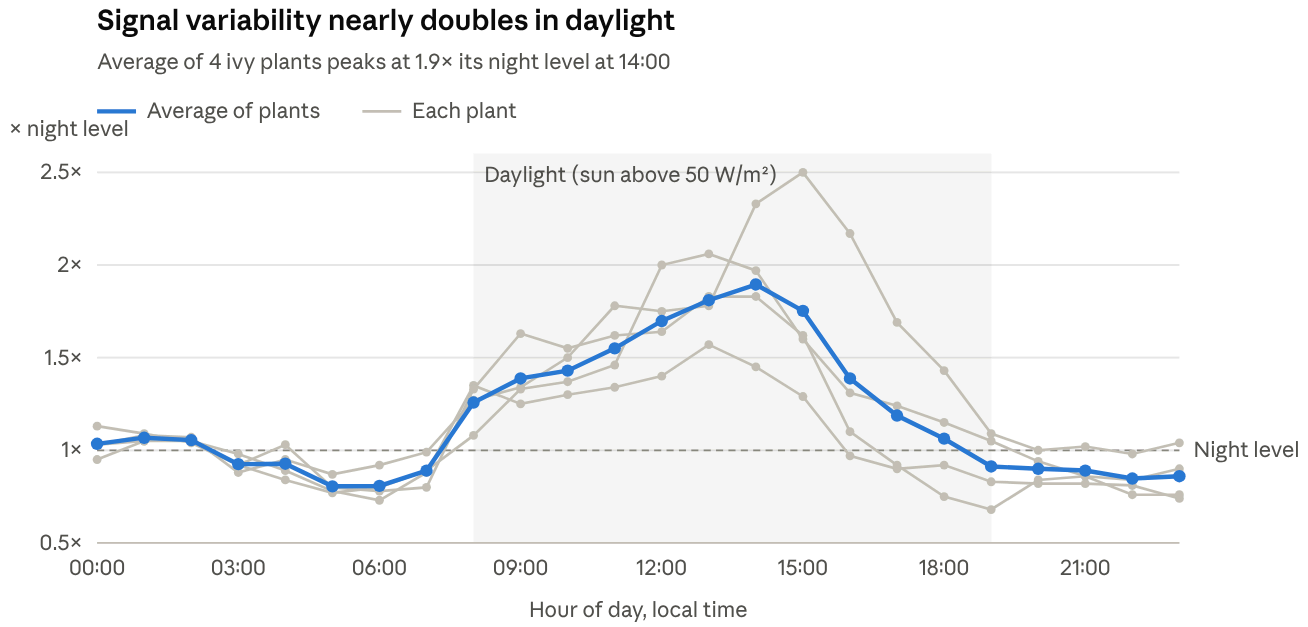
The University of Konstanz ivy dataset (Buss, Aust and Hamann, CC-BY 4.0) has four *Hedera helix* plants recording outdoors from July to November 2024, each with two electrode channels at 1 reading per second. A weather station about 20 m away logged temperature, humidity, solar radiation, wind and rain every 10 minutes.

- **What was used:** plant-days with at least 80% of readings, the same rule as the original paper. That gives 216 plant-days, matching the paper, including 40 days where all four plants recorded at once.

- **Two measures per 10 minutes:** the signal's *level* (its average) and its *variability* (how much the 1-second readings jump around within those 10 minutes).
- **Honest testing:** every score is on days or plants held out of fitting. Slow electrode drift is removed first so it can't be credited to weather or to the plant.

The raw values are in converter units, since the hardware's conversion to millivolts isn't published. Every result below is a ratio, correlation or score, so units don't affect it.

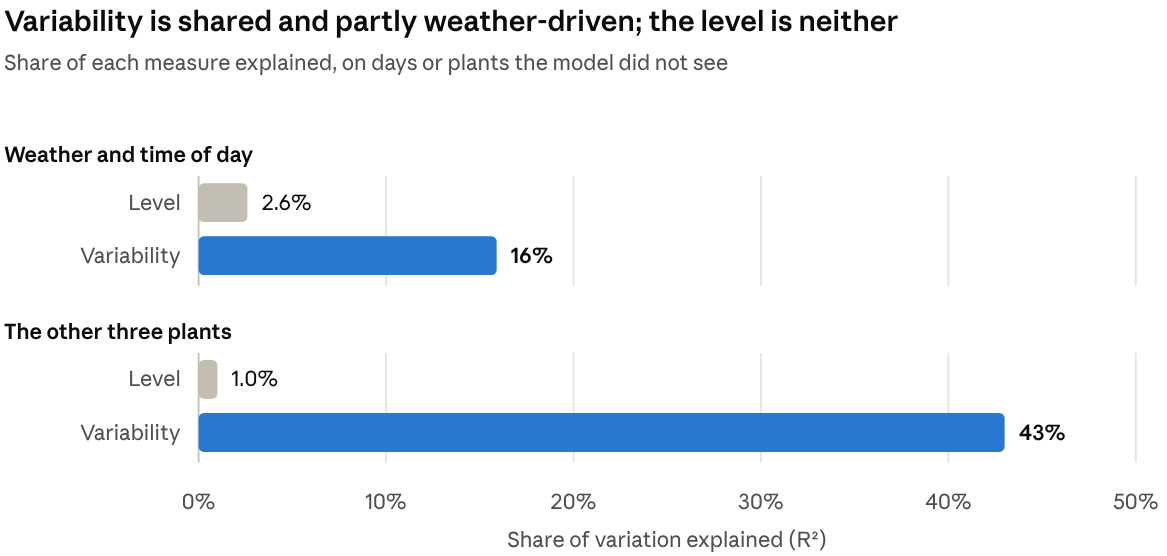
Finding 1: variability follows the sun



Konstanz ivy dataset (Zenodo 15095523) · 4 plants, 216 plant-days, Jul–Nov 2024 · median within-10-minute variability by hour, both channels, relative to 00:00–05:00

All four plants rise from about 08:00 and fall back by evening, in step with sunlight. P1 stays high longest into the afternoon. The signal's level shows no shared daily rhythm like this, which the next chart puts in numbers.

Finding 2: what plants share is variability, not level



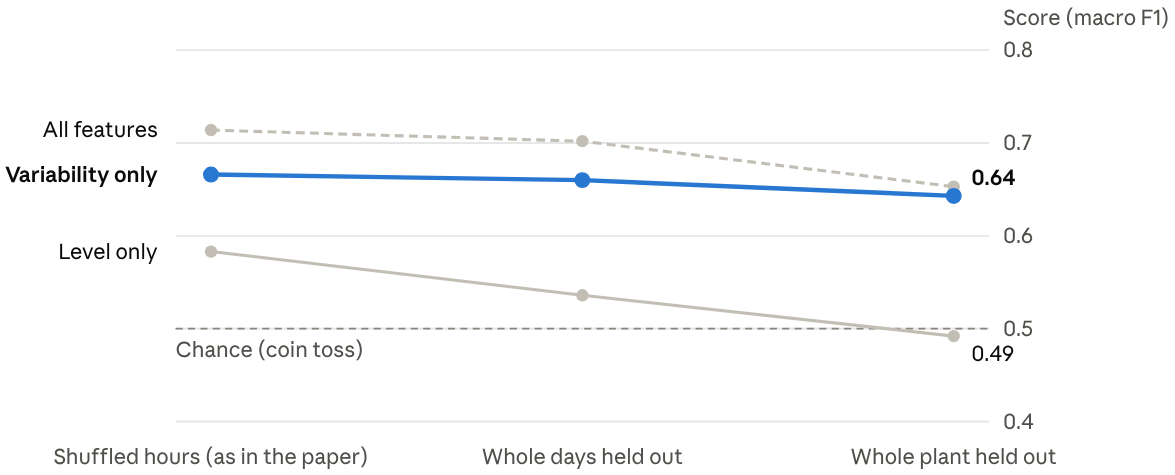
Konstanz ivy dataset · averaged over 4 plants × 2 channels · weather: linear model on 10-minute values, whole days held out · other plants: squared correlation with the other three plants' average, hourly values · slow drift removed in both

The signal's level is close to unpredictable: on average, neither the weather nor the other plants explain more than 3% of it (11% at best for any single plant). Its variability is different. The other three plants, recorded at the same hour, explain 43% of a plant's variability, and weather alone explains 16% of it on unseen days. A nonlinear model (gradient boosting) adds little for variability (17%) and does worse than guessing for level, a sign there is little there to learn.

Finding 3: only variability carries over to a new plant

On a new plant, the level is a coin toss; variability still works

Telling day from night with 1-hour windows, scored three ways from easiest to hardest



Konstanz ivy dataset · 5,162 one-hour windows, day = sun above 50 W/m² · random forest, 10 shuffled splits / 5 day-grouped folds / leave-one-plant-out · features from 10-minute summaries

The original paper scored its models on shuffled 1-hour windows, so hours from the same day sit in both training and test sets. With these simpler features that costs little: holding out whole days lowers scores by 0.01 or less for variability and all features, and by 0.05 for level. The real drop comes on a plant the model has never seen. There, the level falls to chance (0.49), while variability holds at 0.64.

That is the property a cross-species model needs. Whatever transfers to a new tree will look like the signal's dynamics, not its baseline.

Rain and wind: plant or sensor?

Rain is the second-biggest driver of variability after sunlight. During rainy night hours, variability was about 3.5 times that of dry nights (77 rainy hours). By day it was about 2.2 times (36 rainy hours). Wind added nothing once sunlight and rain were accounted for.

In a joint model, sunlight (standardised effect 0.45) and rain (0.37) dominate, and together with temperature, air dryness (vapour pressure deficit) and wind they explain 30% of hour-to-hour variability.

This data can't say whether rain changes the plant or the measurement. Water uptake after rain is a real physiological response, but wet bark and electrodes, splashing and leakage paths also add noise. Someone who has worked outdoors with crops would want this pinned down before any rain-day reading reaches a grower. The same goes for daytime: the Konstanz sensors run on solar power, so part of the daytime rise could come from the device rather than the plant.

What this suggests for a trunk-mounted probe

1. **Read the dynamics, not the level.** The level drifts and differs from plant to plant. Its moment-to-moment variability is what follows the environment and carries over to a new plant.
2. **Judge each reading against its expected weather response.** Removing the weather's effect cuts how closely the plants' variability agrees from a correlation of about 0.44 to 0.22–0.30. A stressed tree should show up as a departure from the expected response, which needs weather data alongside every probe.
3. **A couple of probes can capture the shared signal, not each tree.** The other plants explain about 43% of a plant's variability, so a few probes track site-wide conditions. They won't catch one tree going wrong on its own. Sap flow studies point the same way: Moon et al. needed more than 10 trees to estimate stand water use within 10%.
4. **Log rain and installation details from day one.** Rain shifts variability more than anything except sunlight. Without rain, wetting and sensor-placement records in the field data, a model can't learn to separate plant from sensor.

Research context

The ivy results fit what tree and crop studies report: a strong daily rhythm tied to water movement, stress showing up as a change in that rhythm, and large tree-to-tree differences.

Study	Plants	What it found
Gibert et al. 2006	Poplar trunk, France	Daily swings of about 30–50 mV track sap flow in summer, but not in proportion, and lag it by hours

Study	Plants	What it found
Hao et al. 2021	Poplar, China	Daily swings rise linearly with transpiration and move with xylem water content; sap flow alone isn't the cause
Tran et al. 2019	Greenhouse tomato, Switzerland (Vivent)	Water deficit lowers the baseline and shrinks the daily swing; day vs night classified at 94.6%
Najdenovska et al. 2021	Greenhouse tomato (Vivent)	Spider mite stress detected at about 80% in daylight but only about 65% at night
Buss et al. 2025	Outdoor ivy, Germany (this dataset)	Day vs night, rain, temperature and wind classified at 85–95% F1 from 1-hour windows with shuffled splits
Moon et al. 2015	Korean pine stand	Sap flow varies 34% between trees; one tree gives about 56% error in stand water use, and more than 10 are needed for under 10%
Liu et al. 2024	Larch plantation, China	At least six trees needed to estimate stand sap flow within 10%

The Vivent tomato data is not public, citing commercial interests, so the ivy data is the main open outdoor benchmark.

Limits and next steps

Limits. Four ivy plants at one site in one season is a small, unlike-an-orchard sample. Electrodes in ivy stems aren't Electric Plant's cambium screws in tree trunks. These results use 10-minute summaries of the 1-second data, so they understate what richer features can do: the original paper's 85–95% scores came from 1-second data and over 700 features per hour.

Next steps, in order of value:

1. **Rerun the held-out-plant test on the 1-second data** with richer features, to see how much of the paper's 94% day vs night survives on an unseen plant.

2. **Add US tree data.** SAPFLUXNET has sap flow for 2,714 plants in 202 datasets, many in the US, with radiation and air dryness. It can test the "how many trees" question on real trees with real water use.
3. **Add the almond-orchard climate.** The AmeriFlux US-ASM site in California's San Joaquin Valley gives the weather an orchard probe would face, against Konstanz's.
4. **Add an Irish climate check.** Met Éireann hourly station data shows how often rain would disturb readings in a wet maritime climate.
5. **Separate plant from sensor on rain days** with a dead-stem or resistor control electrode beside a live one.

Sources

- Data: Konstanz ivy electrophysiology dataset (CC-BY 4.0)
- Buss, Aust and Hamann 2025, When Plants Respond
- Electric Plant launch coverage, TechCrunch
- Research table above: Gibert et al. 2006, Hao et al. 2021, Tran et al. 2019, Najdenovska et al. 2021, Moon et al. 2015, Liu et al. 2024 (linked in the table)
- Analysis: Python (pandas, scikit-learn); ridge regression and gradient boosting with whole days held out, random forests for the split comparison